



Department of Electrical and Electronics Engineering

Academic Year 2024 – 2025 (Odd Semester)

Degree, Semester & Branch: III Semester B.E. EEE

Course Code & Title: EE3303 Electrical Machines I

Name of the Faculty member (s): Mrs. S. Jeyanthi

Innovative Practice Description

- **Unit / Topic:** Unit I / Introduction to Indian Standard Specifications (ISS) - Role and Significance in Testing
- **Course Outcome:** CO1
- **Unit Outcome:** 1e
- **Activity Chosen:** One-minute paper
- **Justification:**

One-minute paper activity provides a conceptual bridge between successive class periods. Improve the quality of class discussion by having students write briefly about a concept or issue before they begin discussing it.

- **Time Allotted for the Activity:** 5 minutes

- **Details of the Implementation:**

At the end of the class, students were asked to write about the topic discussed in the class. The students expressed the understood content and the content which were not clear in that particular topic. This activity shows whether the students can able to understand the specific topic and their involvement the particular class.

- **CO – PO / PSO mapping:**

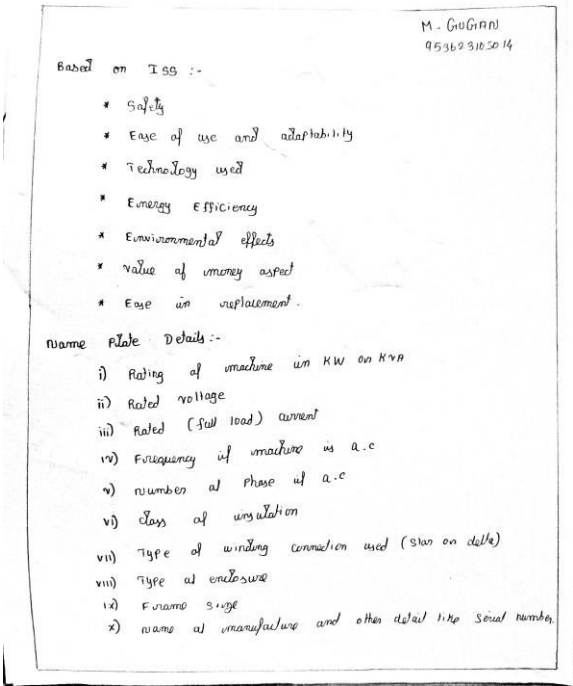
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	1	-	-	-	-	-	-	2	-	2	2	-	-

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO9
	1
Justification for correlation	The students can Function effectively as an individual

• **Images / Screenshot of the practice:**

Innovative Teaching Method Execution	
Introduction to Indian Standard Specifications (ISS) - Role and Significance in Testing – One minute paper	
 Based on ISS :- * Safety * Ease of use and adaptability * Technology used * Energy Efficiency * Environmental effects * Value of money aspect * Ease in replacement Name Plate Details :- - Rating of machine in KW or KVA - Rated voltage - Rated (full load) current - Frequency of machine is a.c - Number of phase of a.c - Class of insulation - Type of winding connection used (star or delta) - Type of enclosure - Frame size - Name of manufacturer and other detail like Serial number	

• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

- ✓ Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- ✓ Students can able to attend the question even in the questions are in indirect form.
- ✓ Students can able to explain the concepts in examination without any confusion.

❖ **Challenges faced in implementation:**

- ✓ Time utilization for conducting activity.

References:

1. I. J. Nagrath and D. P. Kothari, "Electric Machines", McGraw Hill Education, 5th Edition, 2017.
2. P. S. Bimbhra, "Electric Machinery", Khanna Publishers, 2nd Edition, 2021.

Signature of Faculty Member

HOD



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Innovative Practice Description

- **Unit / Topic:** Unit II / Constructional features of D.C. Generator
- **Course Outcome:** CO 2
- **Unit Outcome:** 2a
- **Activity Chosen:** Demonstration
- **Justification:**

The construction detail of the DC Generator is taught using the active learning approach of demonstration, Students' involvement in the learning is improved based on this demonstration approach.

- **Time Allotted for the Activity:** 15 minutes

- **Details of the Implementation:**

After teaching the concept, give students one or two minutes to think about the topic without writing anything.

At the end the Class (Last 15 minutes)

- ✓ I asked the students to think about various parts of the DC machine for 2 minutes.
- ✓ Then I told them to Pair with their neighbours and discuss about the construction of the machine for another 1 minute.
- ✓ Finally, I shown the motor parts for each student and explained.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO2	3	2	1	-	-	-	-	1	-	2	-	2	2	-	2

(1 – Low

2 – Moderate

3 – High)

- PO / PSO mapped:

Innovative practice	PO 9
	1
Justification for correlation	The students can Function effectively as a team

- Images / Screenshot of the practice:



Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

Students told that it is good see all the parts of the machines individually and demonstration helps the students to understand the concepts easily.

❖ *Benefit of the practice:*

1. Students can able to understand the impact of engineering solution on society
2. Students can able to explain the concepts in examination without any confusion.

❖ *Challenges faced in implementation:*

1. Time utilization for conducting activity.

References:

1. I. J. Nagrath and D. P. Kothari, "Electric Machines", McGraw Hill Education, 5th Edition, 2017.
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Name of the Faculty member (s): Mrs. S. Jeyanthi

Innovative Practice Description

- **Unit / Topic:** Unit III / Speed Control of DC motors

- **Course Outcome:** CO 3

- **Topic Learning Outcome:** 3b

- **Activity Chosen:** MATLAB Simulation

- **Justification:**

The various control strategies can be easily implemented and evaluated, providing a cost-effective and time-efficient platform for optimizing speed control performance in DC motors through MATLAB simulations.

- **Time Allotted for the Activity:** 20 minutes

- **Details of the Implementation:**

- Speed control of DC motor were explained and the students were taken to PSS Lab.
- A sample simulation were simulated in the lab and their doubts were clarified
- A Problem was given to the students and asked to simulate individually and their simulation outputs were individually verified.

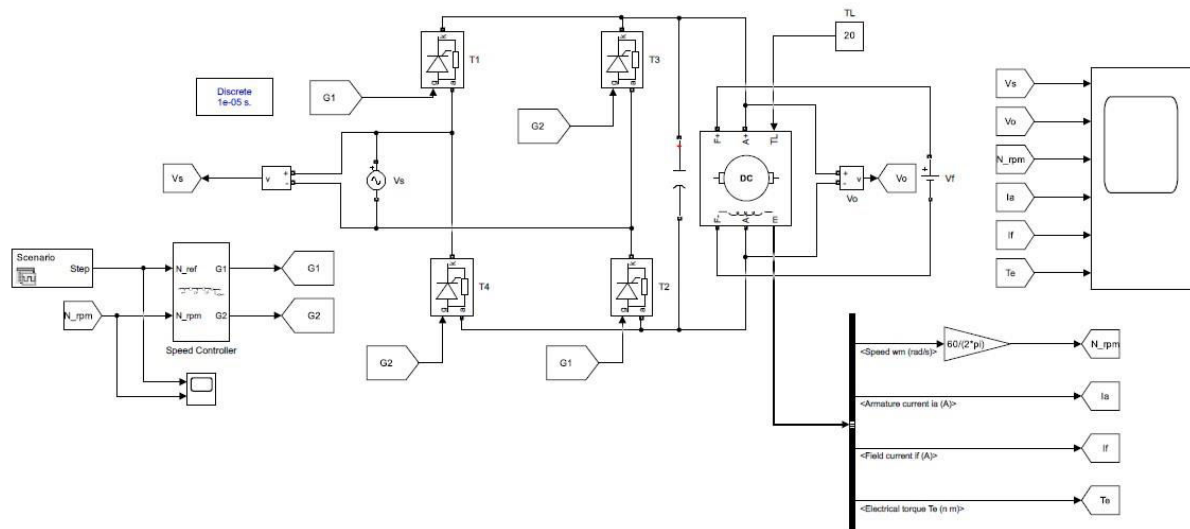
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO3	3	2	1	-	2	-	1	1	-	2	-	2	2	-	2

- **PO / PSO mapped:**

Innovative practice	PO5
	2
Justification for correlation	Student able to analyze the speed control of dc motor using modern software package.

• **Images / Screenshot of the practice:**



• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

- While solving Student felt that their computational time was reduced due to simulation and they were able to solve many problems quickly with the help of simulation.

❖ **Benefit of the practice:**

- Student understood the speed control methods clearly and able to explain the procedure of calculation whenever asked.
- Most of the students have attended the question in their Internal Assessment Test.
- Student Interest towards Matlab Simulation is increased

❖ **Challenges faced in implementation:**

- Initially Few students took more time for their simulation and later by practice their time for completing the simulation is reduced

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Name of the Faculty member (s): Mrs. S. Jeyanthi

Innovative Practice Description

- **Unit / Topic:** Unit IV / Parallel operation of single-phase transformers
- **Course Outcome:** CO 4
- **Unit Outcome:**4b
- **Activity Chosen:** Think pair share
- **Justification:**
 - ✓ It helps students to think individually about a topic or answer to a question.
 - ✓ It teaches students to share ideas with classmates and builds oral communication skills.
 - ✓ It helps focus attention and engage students in comprehending the reading material.
- **Time Allotted for the Activity:** 10 minutes
- **Details of the Implementation:**

Think-Pair-Share innovative practice conducted after explained the concept of Parallel operation of single-phase transformers. First, I asked the students to think about the reason for Parallel operation of single-phase transformers for 2 minutes. Then I make them as a pair to discuss their neighbour's and asked the students to discuss about parameters need to be considered for the Parallel operation of single-phase transformers for 3 minutes. Finally, I asked the one of the team to explain the concept to whole class for further discussion. The students from group share their points and participated in the discussion for 10 minutes.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO4	3	2	1	-	2	-	1	1	-	2	-	2	2	-	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO9
	1
Justification for correlation	The students can Function effectively as a team

• **Images / Screenshot of the practice:**

Innovative Teaching Method Execution
Parallel operation of single-phase transformers
– Think Pair Share

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Parallel operation of Transformers:-

needs of parallel operation:-

- * Increased load.
- * Increased reliability
- * Non availability of large transformer

condition for parallel operation:-

- * The transformer should have same secondary winding
- * The transformer should have same phase sequence.

Types of parallel operation:-

- * full parallel operation of transformers with equal voltage ratio.
- * parallel operation of transformers with unequal voltage ratio.

• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

- ✓ Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

Think-pair-sharing forces all students to attempt an initial response to the question, which they can then clarify and expand as they collaborate. It also gives them a chance to validate their ideas in a small group before mentioning them to the large group, which may help shy students feel more confident participating.

❖ **Challenges faced in implementation:**

I planned the activity for 10 minutes. But in Class room it takes 15 minutes.

References:

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Innovative Practice Description

- **Unit / Topic:** Unit V / Three Phase Transformer Construction
- **Course Outcome:** CO5
- **Topic Learning Outcome:** 5c
- **Activity Chosen:** Field Visit
- **Justification:**

Field visits for the topic of Three Phase Transformer Construction are justified to provide hands-on experience with transformer components, enhance understanding of assembly processes, and observe real-world applications for theoretical knowledge consolidation.

- **Time Allotted for the Activity:** 30 Minutes
- **Details of the Implementation:**

After completing the topic, I took the students to the RIT – Power House and shown them the three phase transformer and explained the construction of transformer.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO5	3	2	1	-	2	-	1	1	-	2	-	2	2	-	1

- **PO / PSO mapped:**

Innovative practice	PO7
	1
Justification for correlation	Due to this activity, students will understand the impact of professional engineering solution for environment. So it is slightly correlated.

- Screenshot of the activity:



Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

Students were interested in attending field visit and requested to conduct more such activities in future.

❖ *Benefit of the practice:*

- ✓ The assessment of effectiveness of the activity was good in a way that while asking questions, most of the students answered and shared many points.
- ✓ After conducting the activity, I understood that the students could be able to recollect the points about substation and grounding methods.

❖ *Challenges faced in implementation:*

- ✓ Time is not enough for this activity, so will allot more time in future.

References:

1. I. J. Nagrath and D. P. Kothari, "Electric Machines", McGraw Hill Education, 5th Edition, 2017.
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